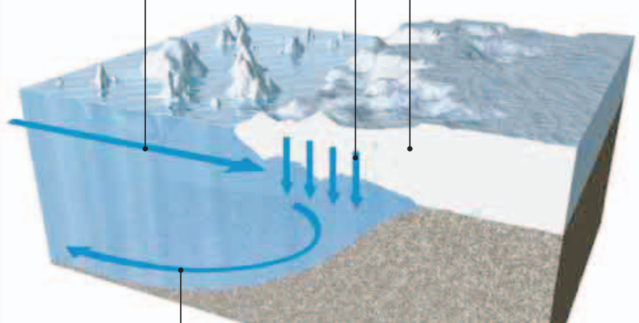
Deepwater currents

The surface currents that swirl around the world's oceans are linked to a network of deepwater currents. These are driven by cool, salty water sinking toward the ocean floor and flowing beneath the warmer surface water, until they eventually come back to the surface. Together, the deepwater currents and surface currents carry ocean water all around the world.

SINKING WATERS

In the polar regions, cold air and floating ice makes the water beneath it very cold. This makes the water molecules move closer together, so the water becomes denser (heavier per liter). Extra salt expelled from the sea ice as it forms makes this cold water even denser and heavier, so it sinks toward the ocean floor.



DEEP CHILL

The coldest deepwater current is called the Antarctic Bottom Water, which flows from beneath the ice that covers the Antarctic Weddell Sea. A similar flow comes from the Ross Sea on the other side of Antarctica. In the north, sinking water near Greenland propels the North Atlantic Deep Water, which flows south to help drive a deepwater current around the globe.

▼ ANTARCTIC ICE
Sea ice forming on the Weddell and Ross seas close to the South Pole makes cold, salty ocean water even colder and saltier, driving powerful deepwater currents.

WOW!
The origin and history of each drop of ocean water can be worked out by analyzing its chemical nature.